

GEMS Installer Manual

Hardware Setup, Network Discovery, Device Integration, and Optimization Strategy Configuration

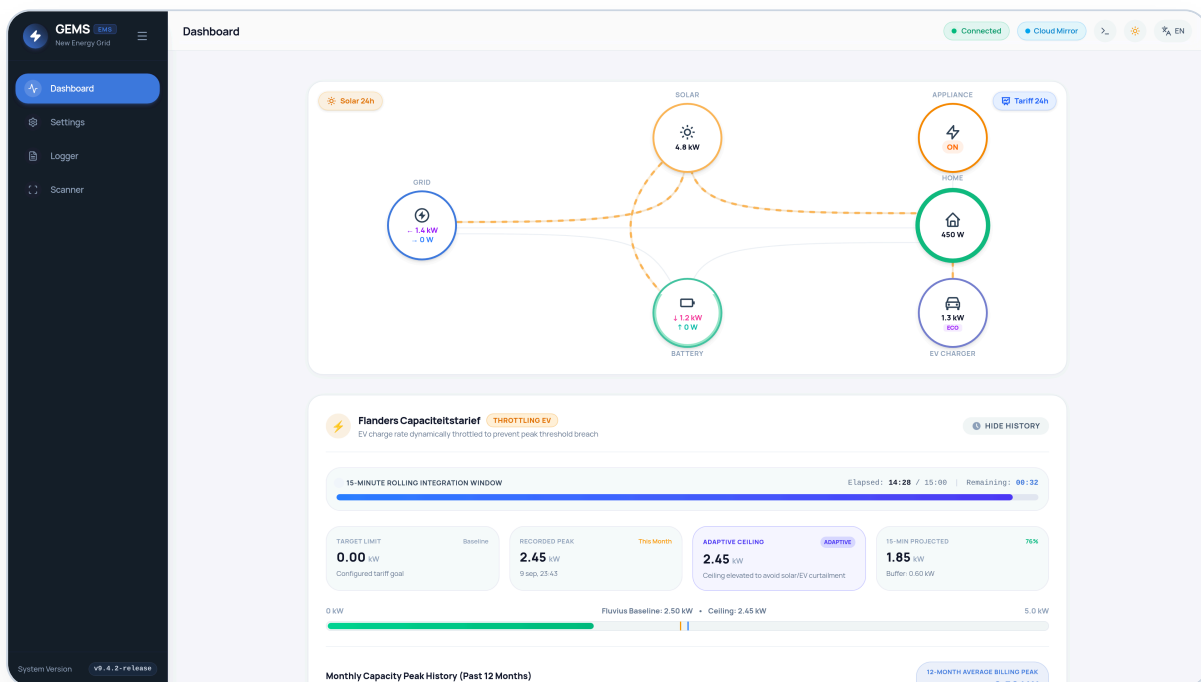
SYSTEM FIRMWARE / BUILD**v9.8.0-release (Official Tag)****TARGET HARDWARE ARCHITECTURE****Raspberry Pi 5 (ARM64) + M.2 NVMe HAT****TARGET AUDIENCE****Electricians, Solar & ESS Installers,
Integrators****REVISION DATE****September 2026**

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1. System Architecture & Hardware BOM

GEMS (Grid Energy Management System) is an autonomous, lightweight, fully UI-driven energy manager developed in Go and Vue 3. It runs directly on the local residential network on a Raspberry Pi without requiring cloud subscription dependencies. To eliminate SD card wear and ensure decades of reliable operation, GEMS runs from an M.2 NVMe SSD using SQLite configured in Write-Ahead Logging (WAL) mode.

Reference Hardware Bill of Materials (BOM)

Component	Specification	Installer Notes
Host Computer	Raspberry Pi 5 (4GB or 8GB RAM, Broadcom BCM2712)	Provides hardware PCIe 2.0 interface. CPU usage stays < 5% during full polling.
Internal Storage	128GB+ M.2 PCIe Gen 3 ×4 NVMe SSD	Low power consumption, durable TLC flash. Write speed up to 600+ MB/s.
M.2 PCIe HAT	Raspberry Pi 5 M.2 PCIe HAT+ or Bottom NVMe Base	Connects to the 16-pin FPC PCIe ribbon connector. Ensure gold pins face correctly.
Power Supply	Official Raspberry Pi 27W USB-C PD (5.1V / 5.0A)	Mandatory to power both the Pi 5 and the NVMe SSD without brownouts.
Cooling Enclosure	Aluminium armor metal case with PWM Active Cooler	Ensures sustained 24/7 operating temperatures below 48°C in electrical cabinets.
Serial Interface	FTDI USB-to-RS485 Isolated Converter or P1 RJ12 cable	Connects to DSMR P1 digital meters or Modbus RTU serial buses (A/B lines).

2. NVMe Bootloader & EEPROM Setup

Brand new Raspberry Pi 5 boards ship with a factory EEPROM configured to boot solely from MicroSD ('BOOT_ORDER=0xf41'). When booting with an M.2 NVMe SSD, the PCIe bus must be enabled via the bootloader.

☀ 3-Minute Quick Fix (Using Raspberry Pi Imager)

1. Insert any MicroSD card into your laptop.
2. Open **Raspberry Pi Imager** → Choose Device: **Raspberry Pi 5** → Choose OS: **Misc utility images** → **Bootloader** → **NVMe/PCIe Boot**.
3. Choose the SD card, click **Write**, and insert into the Pi 5.
4. Power on with the official 27W power supply. Within 5 seconds, the green ACT LED will flash rapidly and the HDMI screen will turn **solid green**.
5. Unplug power and discard/remove the MicroSD card. The EEPROM is now permanently set to 'BOOT_ORDER=0xf461' and 'PCIE_PROBE=1'.

3. OS Deployment & System Services

GEMS provides two official deployment methods for installers:

Option A: Pre-built Custom Raspberry Pi OS Image (Recommended)

Download `gems-os-image.img.xz` from the GitHub release. Flash it directly to the M.2 NVMe SSD via an external USB NVMe enclosure using BalenaEtcher or Raspberry Pi Imager. The pre-built image includes:

- **Nginx Proxy:** Routes port 80/443 to the GEMS backend on port 8080 with WebSocket and SSE support.
- **Kiosk Mode:** Minimal Wayland compositor (Wayfire) displaying Chromium in fullscreen pointing to `http://localhost`.
- **Cockpit Web Console:** Available on port `9090` for browser-based root terminal access.
- **Raspberry Pi Connect:** Pre-installed daemon for secure browser-based remote screen viewing.

Option B: Debian Package (.deb) on Existing OS

```
# Download and install the latest ARM64 Debian package
sudo dpkg -i gems_9.8.0_arm64.deb
sudo apt-get install -f

# Verify service status
sudo systemctl status gems.service
```

The package automatically provisions a dedicated system user (`gems`), installs binaries to `/opt/gems`, and enables the systemd service.

4. Remote Access & Diagnostics Console

Technicians can manage the installation without keyboard or monitor:

- **Web UI:** `http://ems.local` or `http://<IP-ADDRESS>` (Default port 80 or 8080).
- **Cockpit Web Terminal:** `https://<IP-ADDRESS>:9090` (Default User: `admin`, Password: `manufacturer`). Offers web terminal, storage SMART metrics, system reboot, and networking.
- **Remote Screen Sharing:** Run `rpi-connect signin` in Cockpit to link the Pi to your Raspberry Pi ID for one-click remote desktop view.

5. Subnet Hardware Discovery (Scanner)

Before configuring devices, use the built-in **Scanner** in the sidebar. The scanner performs a parallel ARP sweep and TCP port scan across the local /24 subnet, matching MAC addresses against known solar and meter vendors (Huawei, SMA, Easton, Shelly, Alfen, Easee).

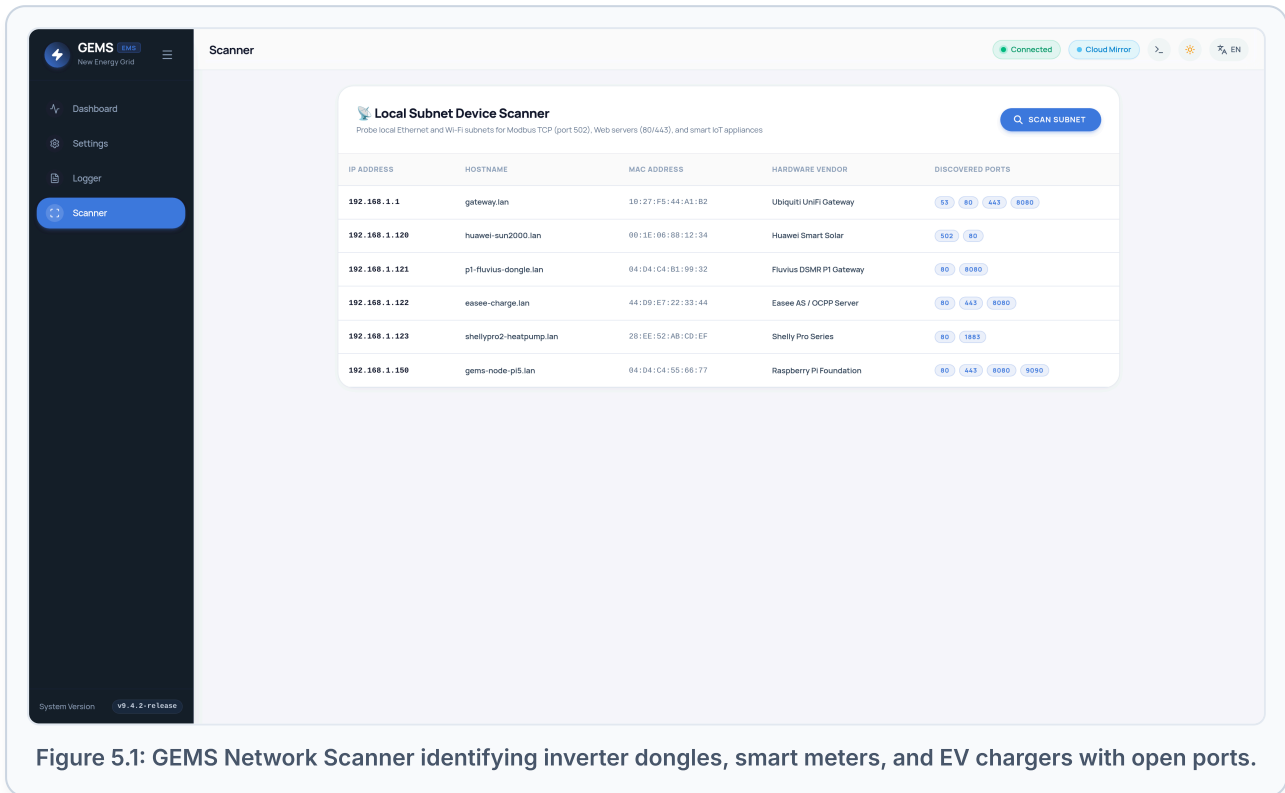


Figure 5.1: GEMS Network Scanner identifying inverter dongles, smart meters, and EV chargers with open ports.

Scanner Usage Steps

1. Click **Scanner** in the primary left navigation.
2. Click the blue **Scan Subnet** button.
3. Review the discovered hosts: note the detected IP, open ports (e.g. 502 for Modbus TCP, 80 for REST), and vendor tags.

6. Adding & Configuring Devices (UI-Driven)

GEMS includes **90 native hardware templates**. Devices are added exclusively through the web interface with zero YAML file editing.

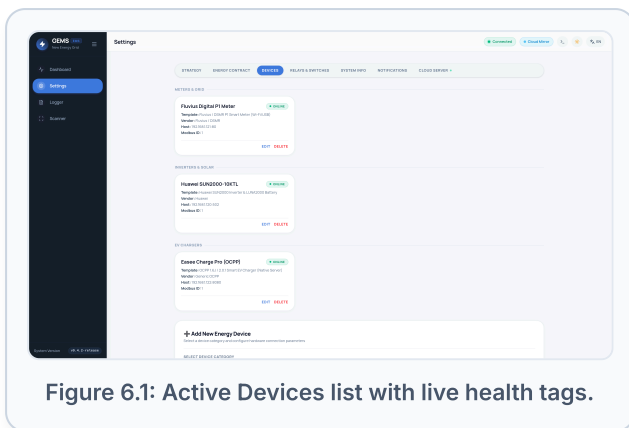


Figure 6.1: Active Devices list with live health tags.

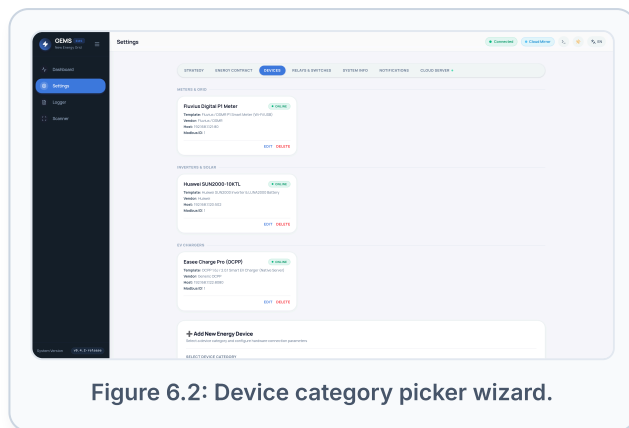


Figure 6.2: Device category picker wizard.

Step-by-Step Onboarding Procedures

A. Solar & Hybrid Inverters (Modbus TCP)

Supported brands include Huawei SUN2000, SMA Sunny Boy / Tripower, Solis 5G/S6, Fronius Gen24, SolarEdge, Sungrow, GoodWe, and Victron GX.

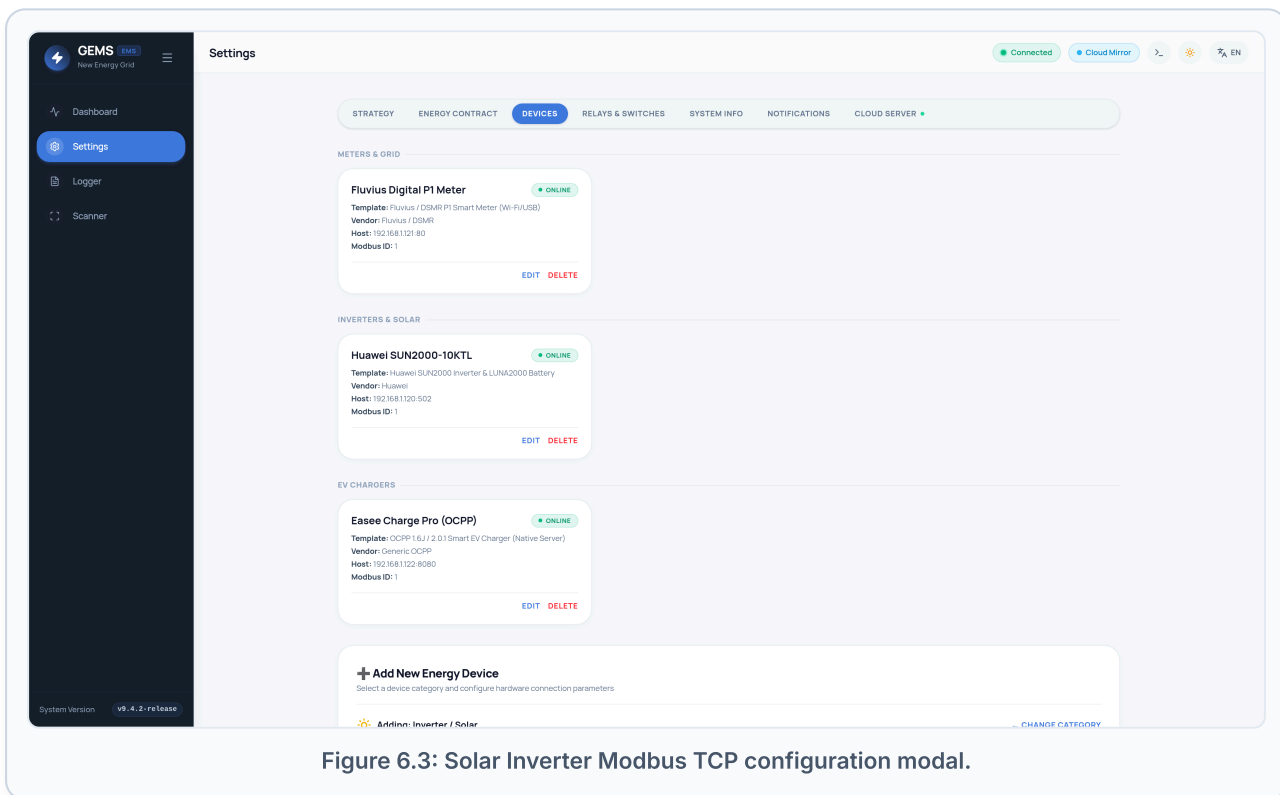


Figure 6.3: Solar Inverter Modbus TCP configuration modal.

Field	Typical Value	Description
Template	Huawei SUN2000 / SMA / Solis	Selects the pre-mapped SunSpec/Modbus register map.
Host IP	192.168.1.120	IP address of the Wi-Fi/Ethernet dongle (e.g. Huawei SDongleA-05).
Port	502 or 1502	Modbus TCP port. Huawei uses 502; some third-party bridges use 1502.
Modbus Slave ID	1 (Huawei/SMA), 126 (SolarEdge)	Slave ID configured on the inverter internal bus.
Rated Power (kW)	e.g. 10.0	Maximum rated AC output power of the inverter.
Has Battery	Checked (if LUNA2000/BYD present)	Enables battery telemetry, SOC tracking, and charge/discharge control.

B. Grid Smart Meters (P1 / Modbus)

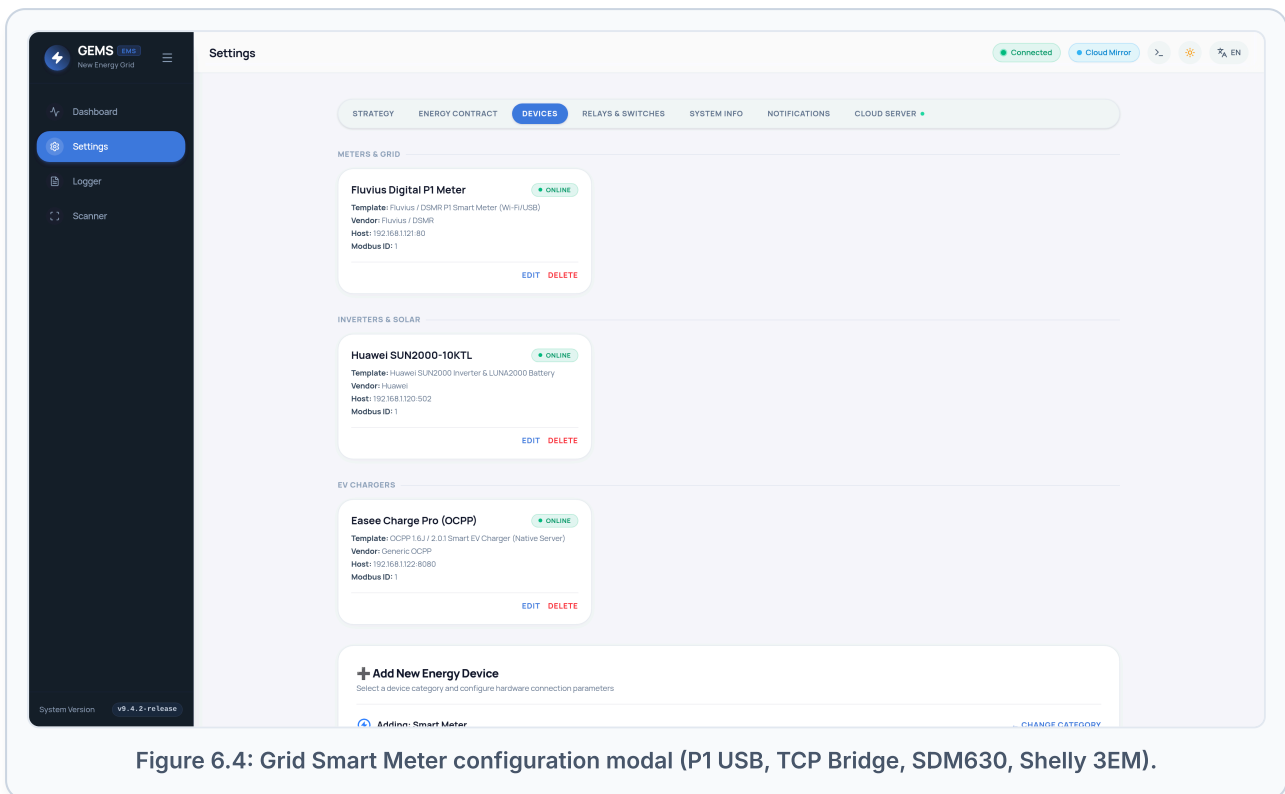


Figure 6.4: Grid Smart Meter configuration modal (P1 USB, TCP Bridge, SDM630, Shelly 3EM).

- **Fluvius / DSMR P1 Serial:** Set host to `/dev/ttyUSB0` , Baud rate `115200` (DSMR 5.0).
- **P1 Wi-Fi Dongle (HomeWizard / TCP Bridge):** Set host to dongle IP (e.g. `192.168.1.121`), port `80` or `8080` .
- **Eastron SDM630 (Modbus):** Connect RS485 A/B lines to USB converter, select template `Eastron SDM630 3-Phase` , Slave ID `1` .

C. EV Charging Stations (Modbus TCP & Native OCPP)

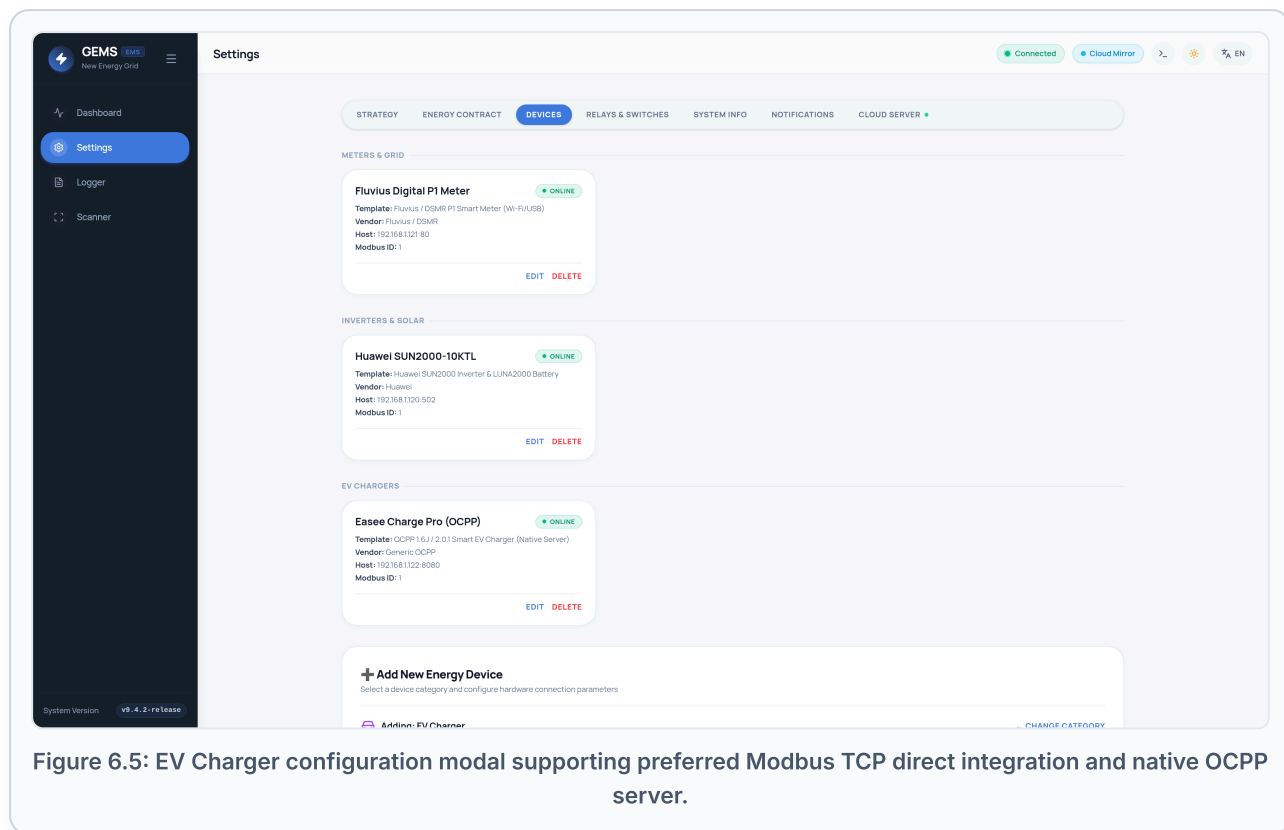


Figure 6.5: EV Charger configuration modal supporting preferred Modbus TCP direct integration and native OCPP server.

⚡ Modbus TCP Integration (Recommended & Preferred)

Modbus TCP is the preferred connection method for EV chargers. Connecting via Modbus TCP (Port 502) offers critical advantages over OCPP:

- **Sub-Second Setpoint Control:** Direct register writes allow instantaneous charge throttling (6A to 32A), essential for reactive Flanders capacity peak shaving and real-time solar curtailment tracking.
- **Simultaneous CPO / Billing Coexistence:** If the charger is already connected to an external Charge Point Operator (CPO) or employer billing reimbursement backend (e.g. E-Flux, Road, Optimile, Easee Cloud) via OCPP, GEMS controls the charger locally over Modbus TCP *without disconnecting or interfering with the cloud billing service*.
- **Zero WebSocket Latency:** Deterministic master-slave fieldbus polling on the local LAN with zero external broker dependencies.

Setup Steps: In your EV charger's web interface or installer tool, enable *Modbus TCP* (default port 502) and note the Unit/Slave ID (typically 1). In GEMS, click **Add Device** → **EV Charger**, choose your hardware template (e.g. *Aifen Eve*, *Keba KeContact P30*, *Mennekes Amtron*, *Webasto Live/Next*, *ABB Terra AC*, *Raedian Gemini/Neo*, *Phoenix Contact Charx*, *Veton*, *SolarEdge Home EVSE*, or *Generic Modbus TCP Charger*), enter the charger's local IP address, Port 502, and Slave ID.

Native OCPP 1.6-J / 2.0.1 WebSocket Server (Alternative)

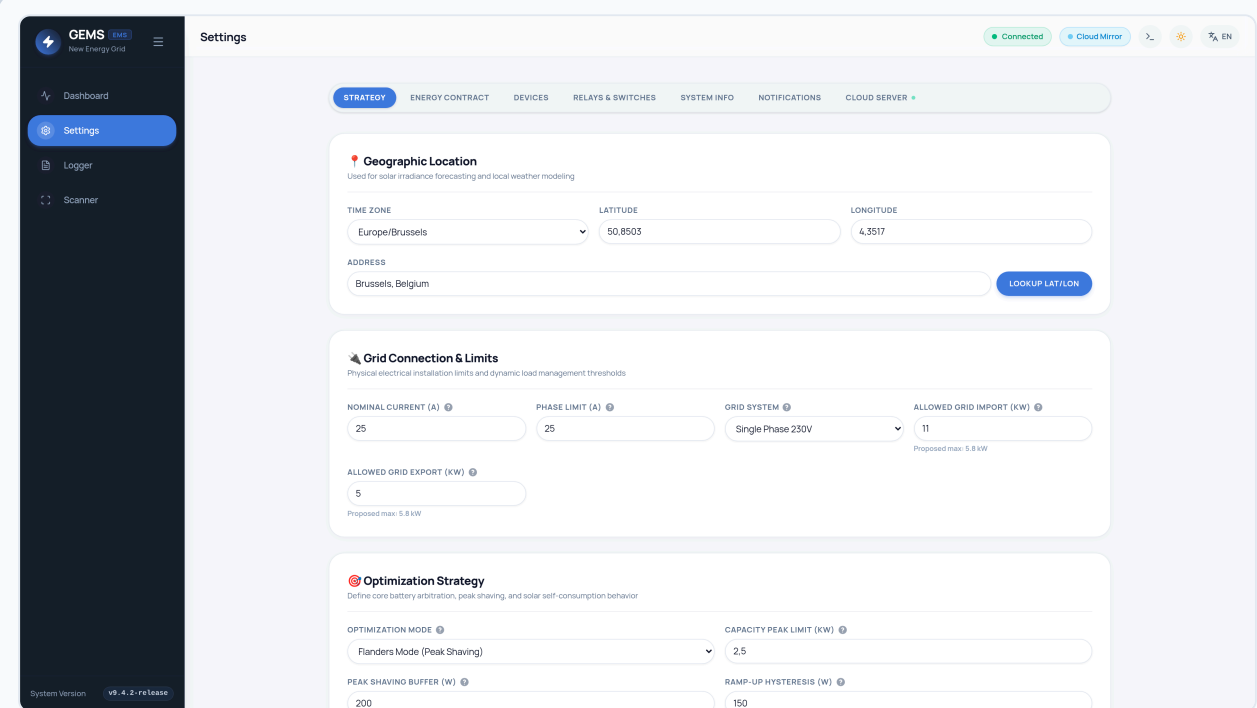
For chargers without Modbus TCP capability or installations without third-party CPO billing backends, GEMS runs an embedded WebSocket OCPP server on port **8887**:

```
Central System URL: ws://<GEMS-IP>:8887/<ChargePointID>  
Example: ws://192.168.1.150:8887/CP001
```

In GEMS: **Add Device** → **EV Charger** → **Template: OCPP 1.6J / 2.0.1 Smart EV Charger**. Enter the matching **ChargePointID**. Once the charger establishes the WebSocket handshake, status turns to **Online**.

7. Grid Connection & Multi-Phase Setup

Navigate to **Settings** → **Strategy** → **Grid Connection & Limits** to specify electrical supply constraints:



The screenshot displays the GEMS Settings interface. The left sidebar shows the navigation menu with 'Settings' selected. The main content area is titled 'Settings' and features a tabbed interface with 'STRATEGY' selected. The 'STRATEGY' tab contains three sections: 'Geographic Location', 'Grid Connection & Limits', and 'Optimization Strategy'. The 'Geographic Location' section includes fields for 'TIME ZONE' (set to 'Europe/Brussels'), 'LATITUDE' (50.8503), 'LONGITUDE' (4.3517), and 'ADDRESS' (Brussels, Belgium), with a 'LOOKUP LAT/LON' button. The 'Grid Connection & Limits' section includes fields for 'NOMINAL CURRENT (A)' (25), 'PHASE LIMIT (A)' (25), 'GRID SYSTEM' (Single Phase 230V), 'ALLOWED GRID IMPORT (KW)' (11), and 'ALLOWED GRID EXPORT (KW)' (5). The 'Optimization Strategy' section includes fields for 'OPTIMIZATION MODE' (Flanders Mode (Peak Shaving)), 'CAPACITY PEAK LIMIT (KW)' (2.5), 'PEAK SHAVING BUFFER (W)' (200), and 'RAMP-UP HYSTERESIS (W)' (150). The bottom of the interface shows the 'System Version' as 'v9.4.2-release'.

Figure 7.1: Grid connection parameters, phase limit amps, and maximum import/export allowance.

Parameter	Format	Field Explanation
Grid System	Dropdown	Single Phase 230V , 3-Phase 400V+N , Or 3-Phase 230V Delta (No Neutral) .
Nominal Current (A)	Float (Amps)	Main circuit breaker rating (e.g. 25A , 32A , 40A , 63A).
Phase Limit (A)	Float (Amps)	Hard safety ceiling per phase. The EMS throttles EV charging if any phase exceeds this limit.
Allowed Grid Import (kW)	Float (kW)	Maximum instantaneous import power allowed by the connection.
Allowed Grid Export (kW)	Float (kW)	Maximum grid feed-in allowed by DNO contract. Set to 0.0 for strict zero-export.

8. Dynamic Contracts & 24h Tariff Curve

Under **Settings** → **Energy Contract**, configure the customer's utility contract. GEMS fetches hourly EPEX Spot Day-Ahead prices daily at 13:00 CET and calculates true retail prices including supplier markup, DNO distribution tariffs, and taxes.

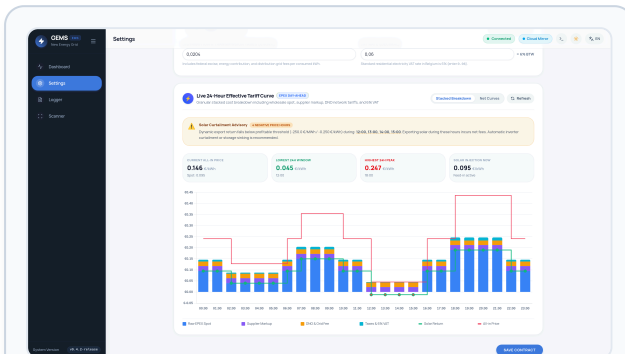


Figure 8.1: Belgian supplier contract presets and tax formulas.

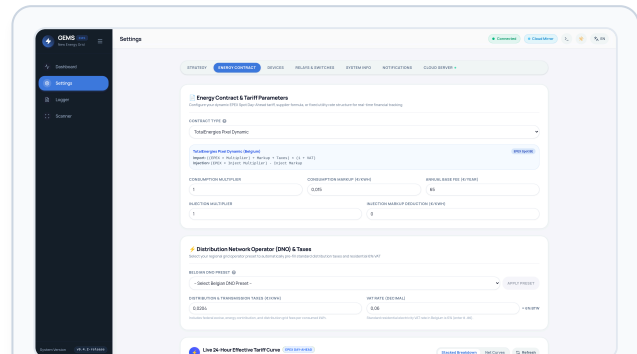


Figure 8.2: 24-Hour effective retail tariff breakdown visualizer.

Supported Contract Models & Supplier Presets

- **Dynamic EPEX Spot:** TotalEnergies Pixel Dynamic, Mega Smart/Cosy Dynamic, Bolt Dynamisch, Engie Dynamic / Flextime, Luminus, Eneco, Frank Energie.
- **DNO Network Tariff Presets:** Fluvius (Flanders), SIBELGA (Brussels), ORES / RESA (Wallonia).
- **Fixed & Dual-Tariff (Peak / Off-Peak):** Configurable peak kWh, off-peak kWh, and fixed injection tariff.
- **Taxes & VAT:** Configurable federal excise duty (default €0.0204/kWh) and Belgian 6% residential VAT.

9. Optimization Strategy & Flemish GSC Protection

Navigate to **Settings** → **Strategy** to configure peak shaving, active inverter curtailment, and green certificate value protection.

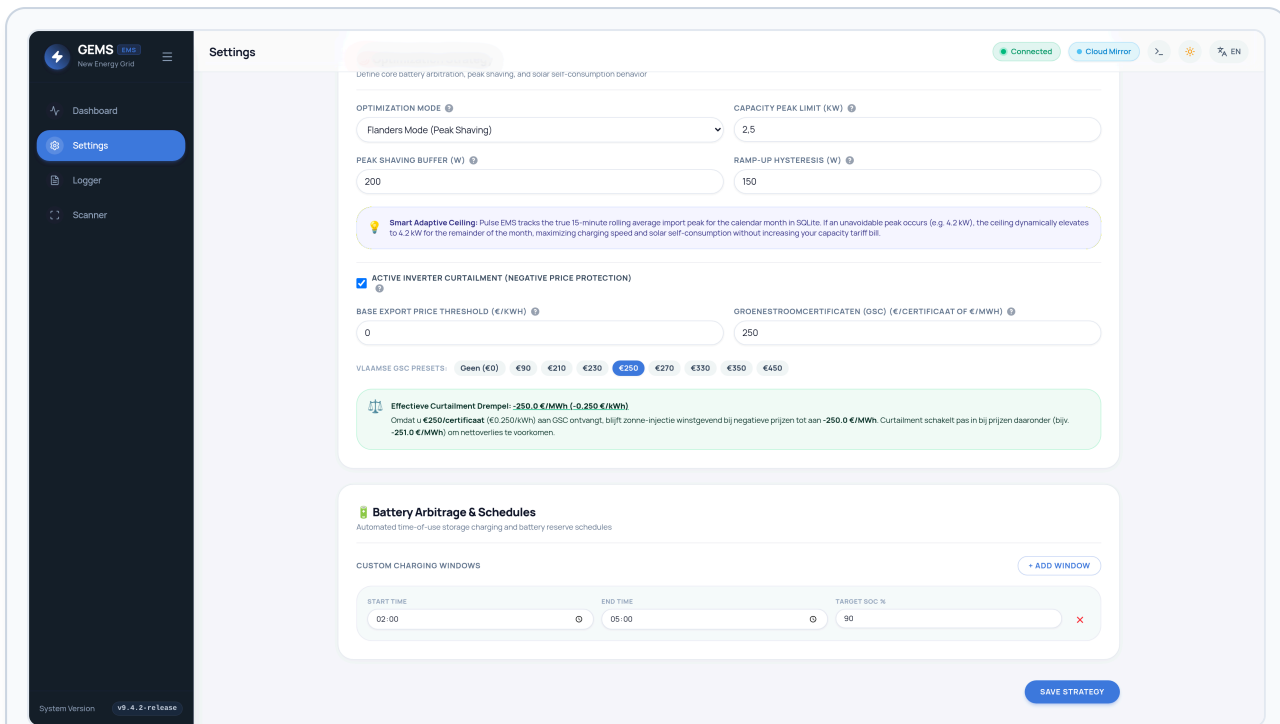


Figure 9.1: Flanders Peak Shaving, Smart Adaptive Ceiling, and Flemish GSC Value presets.

Flanders Capacity Tariff (Capaciteitstarief)

In Flanders, grid distribution fees are calculated based on the homeowner's monthly peak 15-minute average import power (minimum baseline 2.5 kW).

- `capacity_peak_limit_kw`: Hard billing ceiling (default **2.50 kW**).
- `peak_shaving_buffer_w`: Safety margin (default **200 W**) to account for appliance startup spikes.
- `peak_shaving_rampup_w`: Re-acceleration hysteresis rate (default **150 W**).
- **Smart Adaptive Ceiling**: If a unavoidable household peak occurs (e.g. 4.2 kW from oven + heat pump), GEMS dynamically adapts its operational ceiling to 4.2 kW for the remainder of that calendar month. This allows maximum EV charging speed and solar self-consumption without raising the monthly electricity bill!

Active Inverter Curtailment & Groenestroomcertificaten (GSC) Value Protection

During negative spot price hours, exporting solar energy can result in financial penalties. However, legacy Flemish PV systems receive guaranteed **Groenestroomcertificaten (GSC)** (e.g. €250, €350, or €450 per 1,000 kWh).

■ ■ How GSC Protection Prevents Financial Loss

If a customer receives **€250/certificaat (€0.250/kWh)**, solar injection remains profitable even if the market injection price drops to **-€0.20/kWh (-200 €/MWh)** because the homeowner still nets + €0.05/kWh!

Effective Curtailment Threshold = $\text{min_profitable_export_price} - (\text{gsc_value_eur_mwh} / 1000.0)$
Example: $€0.00 - (€250 / 1000) = -0.250 \text{ €/kWh } (-250.0 \text{ €/MWh})$

GEMS will **NOT** curtail the inverter during moderately negative hours. Curtailment only engages if the negative price drops below -250.0 €/MWh (e.g. at -251.0 €/MWh), maximizing the customer's total subsidy earnings!

10. Smart Relays & Heat Pump SG-Ready Automation

GEMS integrates Shelly Pro 2 and Shelly Plus 1PM contactors to automate thermal loads (heat pumps, hot water boilers) based on real-time solar surplus or negative spot pricing.

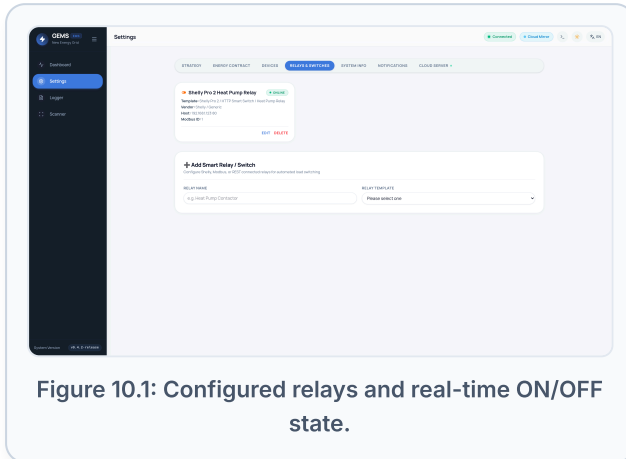


Figure 10.1: Configured relays and real-time ON/OFF state.

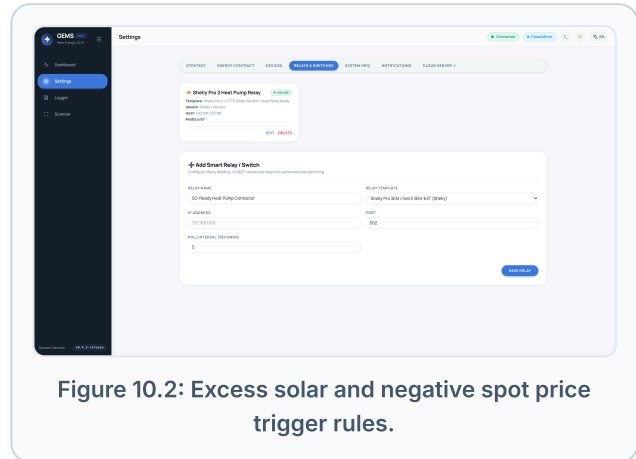


Figure 10.2: Excess solar and negative spot price trigger rules.

Typical Automation Trigger Scenarios

Rule Type	Trigger Condition	Action	Hysteresis	Application
excess_solar	Surplus > 1,200 W	Turn ON	10 Minutes	SG-Ready Heat Pump thermal boost to elevate buffer tank temperature.
negative_spot_price	Tariff ≤ €0.02/kWh	Turn ON	15 Minutes	Domestic hot water boiler electrical resistance element heating.

11. Alerts, Webhooks & Automated Reports

Configure external monitoring under **Settings** → **Notifications**:

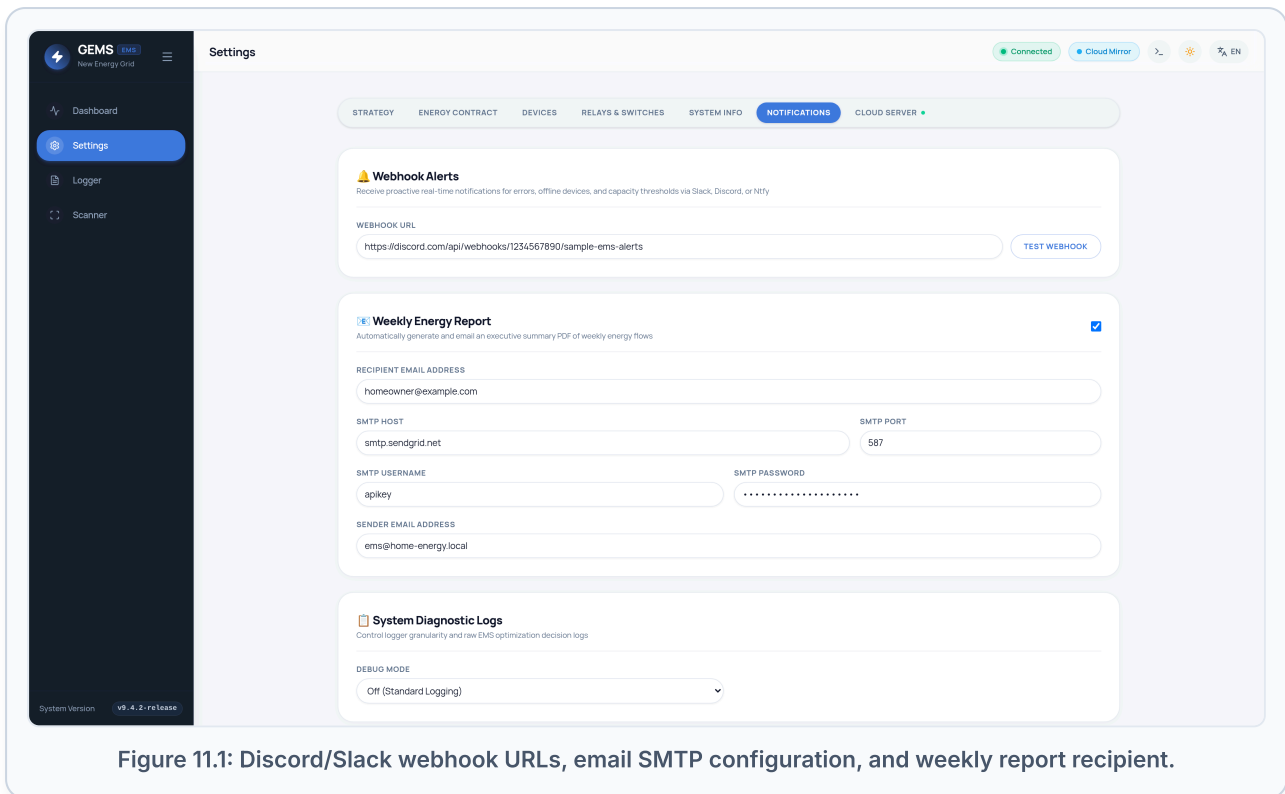


Figure 11.1: Discord/Slack webhook URLs, email SMTP configuration, and weekly report recipient.

- **Discord / Slack / Teams Webhooks:** Real-time alerts on consecutive device polling failures, Modbus communication timeouts, and monthly capacity peak ceiling breaches.
- **Weekly Homeowner Email:** Automatic delivery of a weekly performance summary with total solar generation, self-consumption percentage, and peak grid import.

12. System Diagnostics, Logger & OTA Updates

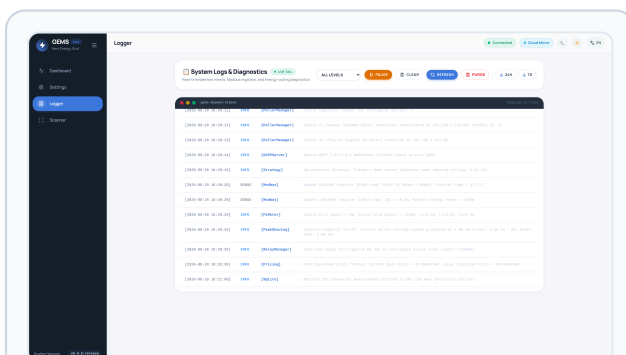


Figure 12.1: Real-time system diagnostics logger.

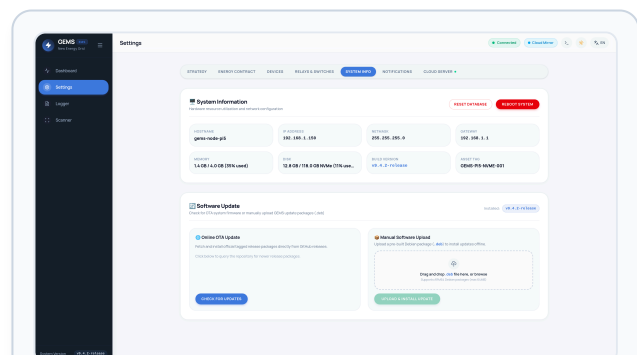


Figure 12.2: NVMe SSD health, build version, and OTA update tool.

Field Commissioning Checklist

- [] Inverter Modbus TCP responding with valid solar wattage on the Dashboard.
- [] Smart meter (P1 or SDM630) reporting import/export power consistent with utility meter.

- [] EV Charger connected via Modbus TCP (preferred) or native OCPP WebSocket (`ws://<IP>:8887/<CP-ID>`) and reporting live current setpoint.
- [] Flanders Capacity Tariff limit set according to customer agreement (default 2.50 kW).
- [] GSC certificate value configured if the installation is eligible for Flemish green certificates.
- [] Energy contract provider and distribution DNO selected under Energy Contract tab.
- [] Weekly report email address tested and saved.